



Movies & TV Shows — Data Analytics Project

*Exploring content trends, countries, genres, ratings and durations
using Python-based data analytics*

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Roll No: 4505/26

Project Overview



Objective

To explore and analyze the Netflix content library and uncover useful patterns and trends using Python-based data analytics.



Dataset

netflix_titles.csv

- Title & content type (Movie / TV Show)
- Country of production
- Release year
- Content rating
- Director & duration
- Genre (listed_in)



Tools Used

- **Pandas**
Data loading, cleaning & analysis
- **NumPy**
Numerical calculations
- **Matplotlib**
Data visualization

Data Preparation & Cleaning

Preparing reliable data before analysis

1

Loaded the Netflix CSV dataset into Python

2

Examined dataset structure & column information

3

Checked for duplicate records

4

Checked for missing values in key columns

5

Handled missing data where necessary

6

Prepared genre & duration columns for analysis

Why it matters: Cleaning reduces errors and inconsistencies so visualizations and conclusions rest on better-quality data.

Analyses Performed in the Project

Eight analyses were carried out on the Netflix dataset:

1

Movies vs TV Shows

Compares the number of Movies and TV Shows in the dataset to see which content type is more common on Netflix.

2

Content Released by Year

Looks at how the number of titles added has changed across years, showing the platform's content growth over time.

3

Top 10 Countries

Identifies the countries contributing the most titles, highlighting Netflix's international content spread.

4

Most Common Ratings

Examines which content ratings (e.g. TV-MA, PG-13) occur most frequently across the catalog.

5

Top 10 Genres

Finds the most frequently listed genres to understand what type of content dominates the platform.

6

Movies vs TV Shows Over Years

Tracks how the release pattern of Movies and TV Shows has shifted across different years.

7

Average Movie Duration

Calculates the typical movie length and studies how durations vary across the catalog.

8

Top 10 Directors

Finds the directors associated with the highest number of titles available on Netflix.

Conclusion

This project applied a complete data analytics workflow — cleaning, transforming and analyzing the Netflix dataset. It examined content type, yearly trends, countries, ratings, genres, durations and directors. Together, these analyses give a clear, well-rounded picture of Netflix's content library.

Thank You